

Multiple Choice (10 marks)

1. Kirkwood gaps are observed in the main asteroid belt including at the position(s) where:
 - (a) asteroids would orbit with a period half that of Jupiter
 - (b) asteroids would orbit with a period twice that of Jupiter
 - (c) asteroids would orbit with a period twice that of Mars
 - (d) A and B
2. What are the conditions necessary for a terrestrial planet to have a strong magnetic field?
 - (a) fast rotation only
 - (b) a rocky mantle only
 - (c) a molten metallic core only
 - (d) both a molten metallic core and reasonably fast rotation
3. We were first able to accurately measure the diameter of Pluto from:
 - (a) a New Horizons flyby in the 1990 s
 - (b) Hubble Space Telescope images that resolved Pluto's disk
 - (c) brightness measurements made during mutual eclipses of Pluto and Charon
 - (d) radar observations made by the Arecibo telescope
4. Which is not a similarity between Saturn and Jupiter's atmospheres?
 - (a) a composition dominated by hydrogen and helium
 - (b) the presence of belts zones and storms
 - (c) an equatorial wind speed of more than 900 miles per hour
 - (d) significant "shear" between bands of circulation at different latitudes
5. The terrestrial planet cores contain mostly metal because
 - (a) the entire planets are made mostly of metal.
 - (b) metals condensed first in the solar nebula and the rocks then accreted around them.
 - (c) metals sank to the center during a time when the interiors were molten throughout.
 - (d) radioactivity created metals in the core from the decay of uranium.

True / False (10 marks)

Answer True or False

6. True or False: Sunspots are black regions on the Sun that experience a significant increase in number every 9 years, referred to as the solar cycle.
7. True or False: The lithosphere consists of the softer rocky material found in the mantle beneath the crust.
8. True or False: Observing a star through cool hydrogen gas reveals only emission lines characteristic of the star's composition.
9. True or False: Volcanic heating cannot account for Mercury's lack of a permanent atmosphere due to its insufficient volcanic activity.
10. True or False: Helium is the second most abundant element in the solar system, following hydrogen in terms of prevalence.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the reasons why Saturn's size is comparable to Jupiter's despite having a significantly lower mass, focusing on the effects of mass on density and planetary structure.
12. Discuss the arrangement of electromagnetic radiation categories from shortest to longest wavelength, analyzing their implications in astronomy and how they affect observational techniques.

End of Paper